

RUTUPARNA SATYAJIT PURANDARE

Pune, Maharashtra, India

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PROFILE

Electronics and Telecommunication Engineering undergraduate with hands-on experience designing and building robotics, embedded systems, and hardware-software integrated prototypes. Developed CNC automation systems, Raspberry Pi-based surveillance platforms, and AI-assisted electronic design tools spanning mechanical fabrication, electronics, firmware, and desktop software. Seeking internships in robotics, embedded systems, electronics hardware, and engineering-focused startups.

EDUCATION

Bachelor of Engineering (Electronics & Telecommunication Engineering)

Marathwada Mitra Mandal's College of Engineering (MMCOE), Pune | Expected Graduation: 2029

- Semester 1 SGPA: 8.5
- Semester 2 SGPA: 8.3

Higher Secondary Certificate (HSC)

Late P. Jog Junior College, Pune | Maharashtra State Board | 2025

- MHT-CET: 86.38 Percentile

Secondary School Certificate (SSC)

Abhinava Vidyalaya English Medium High School, Pune | 2023

- SSC 10th Board: 85.80%
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TECHNICAL SKILLS

Programming: C, C++, Python

Embedded & Hardware: Arduino, Raspberry Pi 4B, ESP32, Battery Management Systems (BMS)

Electronics: Oscilloscope, Multimeter, Soldering, Analog Electronics, Common Emitter Amplifier Design

Design & Engineering: KiCad, AutoCAD, CNC Systems, Stepper Motor Control

Software & Tools: Linux, Git, GitHub, OpenCV (Project Exposure), Tkinter

Engineering Areas: Robotics, Embedded Systems, Hardware-Software Integration, Power Systems

PROJECTS

Infinity Sketcher | Lead Developer

GitHub: github.com/Rutu2701/Infinity-Sketcher

- Designed and fabricated a custom 12 × 12 cm CNC drawbot using Arduino, dual NEMA 17 stepper motors, and DRV8825 motor drivers.

- Developed a complete image-to-motion workflow supporting PNG, JPG, and JPEG inputs for automated sketch generation.
- Implemented custom serial communication between a Python desktop application and Arduino-based motion controller.
- Built a Tkinter-based desktop interface for image upload, parameter tuning, and machine control.
- Completed mechanical design, electronics integration, and deployment within a two-week development cycle.
- Published project documentation and source code on GitHub.

Circuit Forge | Developer

Github: github.com/Rutu2701/Circuit-Forge

- Developing an AI-assisted KiCad workflow that generates schematic component layouts from natural-language circuit descriptions.
- Built automated component selection and schematic generation functionality integrated with KiCad workflows.
- Investigating PCB routing automation challenges involving spatial reasoning and component topology constraints.
- Exploring practical applications of AI-assisted Electronic Design Automation (EDA).

Raspberry Pi Surveillance Cam Buggy

- Engineered a fully wireless Raspberry Pi 4B surveillance rover with onboard camera streaming and Bluetooth-based remote control.
- Integrated hardware, software, and mechanical subsystems independently from concept to deployment.
- Achieved approximately 3–4 hours of battery-powered operation and reliable control at a 6-meter line-of-sight range.
- Configured real-time local video streaming for remote monitoring applications.

Manual 3-Axis Agricultural Gantry System

- Led a six-member team in developing a three-axis agricultural gantry prototype for precision irrigation and mechanical weed removal.
- Designed the motion architecture, wiring, and control methodology while coordinating implementation efforts.
- Implemented adjustable vertical positioning and targeted water delivery for individual plant treatment.
- Successfully demonstrated the prototype at a school-level engineering exhibition.

LEADERSHIP & ACTIVITIES

Antrix Rocketry Club | Battery Management System Engineer

- Responsible for power architecture planning and battery selection for rocket avionics and sensor systems.
- Evaluated power requirements and charging strategies for a 2S1P Li-ion battery configuration.

- Contributed to battery safety, runtime estimation, and power distribution planning.

Prism College Magazine | Student Researcher

- Published a technical article as part of the college magazine editorial and research team.
- Collaborating on additional research-oriented publications and technical content.